

9.5.10

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

If the sum of zeroes of the polynomial is $\sqrt{2}$, then find the value of k .

$$p(x) = 2x^2 - k\sqrt{2}x + 1$$

Solution:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1)$$

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}. \quad (2)$$

Substituting the line into the conic gives

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \quad (3)$$

$$\Rightarrow \kappa^2 (\mathbf{m}^\top \mathbf{V} \mathbf{m}) + 2\kappa \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + (\mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f) = 0 \quad (4)$$

$$\kappa_1 + \kappa_2 = -\frac{2\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})}{\mathbf{m}^\top \mathbf{V} \mathbf{m}}. \quad (5)$$

Given: (Roots are given by intersection of conic with x-axis)

$$\mathbf{V} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -\frac{k\sqrt{2}}{2} \\ 0 \end{pmatrix}, \quad f = 1 \quad (6)$$

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (7)$$

$$\mathbf{V} \mathbf{h} + \mathbf{u} = \begin{pmatrix} -\frac{k\sqrt{2}}{2} \\ 0 \end{pmatrix}, \quad (8)$$

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 2. \quad (9)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = -\frac{k\sqrt{2}}{2} \quad (10)$$

$$\kappa_1 + \kappa_2 = -\frac{2(-\frac{k\sqrt{2}}{2})}{2} = \sqrt{2}. \quad (11)$$

$$\Rightarrow \boxed{k = 2}. \quad (12)$$

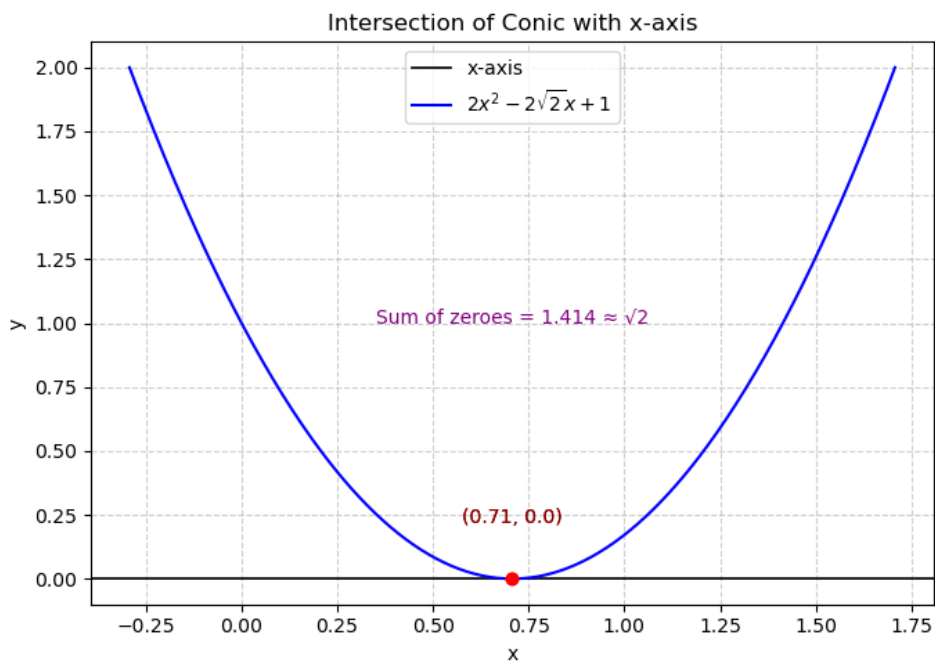


Figure 1: Caption